

# Forkfall

## Player & Operator Guide — Testnet Alpha

Every step from an empty wallet to a settled ranked match: minting the soulbound starter deck, buying and opening packs, building and registering decks, matchmaking, playing a match, and settling it on-chain.

|                 |                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------|
| <b>Version</b>  | MVP 0.1 · branch ccr-63eea11d-n5kv9d                                                         |
| <b>Networks</b> | Base Sepolia (84532, settlement hub) · Robinhood Chain testnet (46630) · local Anvil (31337) |
| <b>Card set</b> | Prototype set: 40 collectible cards (8 per race + 8 neutral)                                 |
| <b>Tools</b>    | Web app, Foundry (forge / cast), agent SDK, MCP server                                       |

**Testnet only.** Every contract reverts if deployed on a chain other than Anvil, Base Sepolia, Robinhood Chain testnet or Ethereum Sepolia, and every Foundry script and the server refuse other chains. Cards, packs and tokens here have no real value. Never use a wallet that holds real funds.

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# 1. Getting set up

Forkfall has three ways in. Most players only need the first; on-chain actions (starter decks, packs, decks, settlement) go through Foundry.

| SURFACE         | USED FOR                                                                                                | WHERE                                          |
|-----------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Web app         | Practice, matchmaking, playing, watching, reading results                                               | http://localhost:8787 (or your operator's URL) |
| Foundry scripts | Minting starters, buying/opening packs, registering decks and agents, settling matches, checking status | contracts/script/Play.s.sol                    |
| Agent SDK / MCP | Bots and AI agents playing through the same API as humans                                               | packages/sdk, apps/mcp                         |

## Requirements

- Node 26 and pnpm 12.
- Foundry 1.8.3 (forge, cast, anvil).
- A **testnet-only** wallet private key with a little Base Sepolia ETH for gas (and for packs, if you pay in ETH).

## Install

```
git clone --recursive https://github.com/gewubuntu/Forkfall.git && cd Forkfall
pnpm install
cp .env.example .env          # fill in TESTNET values only
```

The commands in this guide use these shell variables:

| VARIABLE     | MEANING                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| \$PK         | Your testnet private key (the wallet that plays and owns cards)                                                                    |
| base_sepolia | RPC alias defined in contracts/foundry.toml; reads BASE_SEPOLIA_RPC_URL. Use robinhood_testnet or local instead where relevant.    |
| Book         | contracts/deployments/<chainId>.json, the contract address book written by the deploy script. Every script reads it automatically. |

**All Foundry commands run from the contracts/ folder.** Run `cd contracts` first. Commands that change state need `--broadcast`; leave it off to dry-run.

## Race ids

On-chain calls identify races by number:

| ID | RACE     | HOME CHAIN      |
|----|----------|-----------------|
| 1  | Agents   | Base            |
| 2  | Prophets | Base            |
| 3  | Brokers  | Robinhood Chain |

## 2. Deploying the contracts operators

Players can skip this section. Operators deploy once per network; the deploy script writes the address book that everything else reads.

### Local (Anvil)

```
anvil # terminal 1
pnpm deploy:local # anvil dev key #0 (public, local only)
pnpm server:local
```

### Base Sepolia: the settlement hub

- 1 Create two **testnet-only** keys: a **deployer** (needs about 0.05 Base Sepolia ETH; usually far less is used) and a **referee** (holds REFEREE\_ROLE and co-signs ranked results). Store the deployer in an encrypted keystore:

```
cast wallet import forkfall-deployer --interactive
```

- 2 Copy .env.example to .env and set ETHERSCAN\_API\_KEY (an Etherscan API v2 key verifies on Basescan), REFEREE\_ADDRESS and HOUSE\_PRIVATE\_KEY (the referee key).

- 3 Deploy, verify on Basescan, and run the read-only health check:

```
pnpm deploy:base-sepolia
pnpm check:base-sepolia
```

- 4 Commit contracts/deployments/84532.json so everyone uses the same addresses.

### Robinhood Chain testnet (optional)

```
forge script script/Deploy.s.sol --rpc-url robinhood_testnet --broadcast --account forkfall-deployer
```

Optional environment variables: TREASURY\_ADDRESS (pack proceeds; defaults to the deployer), CARD\_URI, PACK\_PRICE\_WEI (default 0.0001 ETH).

### What gets deployed

| CONTRACT     | ROLE                                                                                              |
|--------------|---------------------------------------------------------------------------------------------------|
| CardRegistry | ERC-1155 card tokens. Id $n$ is the tradeable card; id $10000 + n$ is its soulbound starter twin. |
| StarterDecks | Mints the free soulbound starter deck.                                                            |
| PackSale     | Sells 5-card packs for ETH, test USDC or test token.                                              |
| Crafting     | Scrap tradeable cards, craft any card.                                                            |
| DeckRegistry | Registers 30-card decks and checks ownership and ranked legality.                                 |

|                                  |                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------|
| AgentRegistry /<br>HumanRegistry | ERC-8004 agent identities, badges and bans; optional human verification for season rewards. |
| MatchSettlement                  | Verifies signed results, records wins/losses and per-season Elo.                            |
| SeasonRewards                    | Merkle-based end-of-season reward claims.                                                   |
| TestUSDC / TestFALL              | Faucet tokens (tUSDC, 6 decimals; tFALL, 18 decimals).                                      |

### Run the referee server and web app

```
pnpm web:build
pnpm server          # reads .env: Base Sepolia (84532), RPC, referee key
# → http://localhost:8787 serves the API and the web app
```

**Startup checks.** The server refuses to start unless the RPC is on chain 84532, MatchSettlement is deployed, and HOUSE\_PRIVATE\_KEY holds REFEREE\_ROLE. For UI work before a deployment exists, pnpm server:offchain skips on-chain checks; results from that mode cannot settle.

### 3. Minting the soulbound starter deck

**In the web app.** Open **Collection** → **Free starter decks** and click **Claim**. The Foundry commands below do the same from a terminal.

Every player and agent gets a free 30-card starter deck per race. Nobody has to pay to play, and starter decks are legal in every mode, including ranked.

#### What you get

- Two copies of each of the race's seven non-legendary cards, plus two copies of each of the eight neutral cards: 30 cards.
- They are minted as **soulbound** token ids (10000 + card id). They cannot be transferred, sold or scrapped. That stops farming.
- In decks and in game they count exactly like the normal card.
- One claim per race per wallet. You can claim all four races.

#### Steps

- 1 Mint the starter deck for a race (here Agents = 1):

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "claimStarter(uint8)" 1
```

- 2 Register it as a deck so you can queue ranked. This prints the **deckId**; save it.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "registerStarterDeck(uint8)" 1
```

- 3 Check your collection:

```
forge script script/Play.s.sol --rpc-url base_sepolia --sig "status(address)" <your address>
```

#### Starter deck contents

**Agents** (race id 1): Launch Bot x2, Compute Node x2, Cron Job x2, Sentry Drone x2, Swarm Deployer x2, Firewall x2, Mainframe x2, Cold Wallet x2, Validator x2, Airdrop x2, Liquidator x2, Bridge Runner x2, MEV Searcher x2, Hard Fork x2, Genesis Block x2

[1,1,2,2,3,3,4,4,5,5,6,6,7,7,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

**Prophets** (race id 2): Tea Leaves x2, Seer's Acolyte x2, Star Chart x2, Oracle Guard x2, Called It x2, Market Seer x2, Prophecy of Ruin x2, Cold Wallet x2, Validator x2, Airdrop x2, Liquidator x2, Bridge Runner x2, MEV Searcher x2, Hard Fork x2, Genesis Block x2

[9,9,10,10,11,11,12,12,13,13,14,14,15,15,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

**Brokers** (race id 3): Intern x2, Bond Desk x2, Analyst x2, Index Fund x2, Portfolio Manager x2, Compound Interest x2, Trust Vault x2, Cold Wallet x2, Validator x2, Airdrop x2, Liquidator x2, Bridge Runner x2, MEV Searcher x2, Hard Fork x2, Genesis Block x2

[17,17,18,18,19,19,20,20,21,21,22,22,23,23,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

**Degens** (race id 4): Meme Critter ×2, Paper Hands ×2, Pump Frog ×2, Ape In ×2, Rug Pull ×2, Hype Man ×2, Leverage ×2, Cold Wallet ×2, Validator ×2, Airdrop ×2, Liquidator ×2, Bridge Runner ×2, MEV Searcher ×2, Hard Fork ×2, Genesis Block ×2

[25,25,26,26,27,27,28,28,29,29,30,30,31,31,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]

**Errors you may see.** AlreadyClaimed(player, race): you already minted that race. BadRace: race must be 1-4. Soulbound(id): you tried to transfer a starter copy.

## 4. Buying and opening normal packs

**In the web app.** Open **Collection → Packs**: pick 1-10 packs, buy with ETH or test USDC (the app asks for approval first and has a free tUSDC button), and open sealed packs once they show **Ready to open**. Cards flip one by one; first copies are marked NEW.

Normal packs mint **tradeable** cards. Opening is a two-step process: buying commits to a future block, and opening reads that block's hash to roll the contents. That way neither you nor the seller can pick the result.

### Pack contents (5 cards)

| SLOT | RARITY                                                 |
|------|--------------------------------------------------------|
| 1-3  | Common                                                 |
| 4    | Uncommon                                               |
| 5    | Rare, upgraded to <b>Legendary</b> about 1 in 10 packs |

### Prices (testnet defaults)

| PAY WITH | PRICE PER PACK | HOW TO GET IT                              |
|----------|----------------|--------------------------------------------|
| ETH      | 0.0001 ETH     | Base Sepolia faucet                        |
| tUSDC    | 2 tUSDC        | dripTestUsdc(): 100 tUSDC once per day     |
| tFALL    | 100 tFALL      | drip() on TestFALL via cast: 1,000 per day |

You can buy 1-10 packs per transaction. Proceeds go to the operator's treasury address.

### Buy with ETH

- 1 Buy packs. The script prints the first pack id; ids are consecutive.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "buyPacks(uint256)" 2
```

- 2 Wait at least **2 blocks** (a few seconds on Base Sepolia).

- 3 Open each pack by id. The five minted card ids are printed.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "openPack(uint256)" 0
```

## Buy with test USDC

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "dripTestUsdc()"
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "buyPacksWithTestUsdc(uint256)" 3
```

The script approves the exact amount and buys in one go. Then open each pack as above.

## Buy with tFALL (via cast)

```
B00K=deployments/84532.json
cast send $(jq -r .TestFALL $B00K) "drip()" --rpc-url base_sepolia --private-key $PK
cast send $(jq -r .TestFALL $B00K) "approve(address,uint256)" $(jq -r .PackSale $B00K) 100ether --rpc-url
base_sepolia --private-key $PK
cast send $(jq -r .PackSale $B00K) "buyWithToken(address,uint256)" $(jq -r .TestFALL $B00K) 1 --rpc-url
base_sepolia --private-key $PK
```

**Waited too long?** A block hash is only readable for 256 blocks (about 8.5 minutes on Base). If you open later, the pack is not lost: the call re-commits it to a new block (PackRecommitted event, nothing minted). Wait 2 more blocks and open again.

**Errors.** TooEarly(revealBlock): wait for the reveal block to pass. WrongPayment: ETH sent must equal price × count. NotOwner / AlreadyOpened: wrong pack id. BadCount: buy 1–10 packs.

**Testnet randomness.** The block-hash method is fine for testnet. The design document requires VRF before any mainnet beta.

# 5. Your collection and crafting

**In the web app. Collection → Your cards** shows every card with your copies (♦ = soulbound). Click a card to scrap or craft it; **Scrap extras** burns copies beyond what a deck can use in one transaction.

## Viewing your collection

```
forge script script/Play.s.sol --rpc-url base_sepolia --sig "status(address)" <your address>
```

This lists your season rating, your win/loss/draw record, and every card you own as *card id × playable copies*. Playable copies are tradeable plus soulbound copies combined.

## Trading

Tradeable cards (ids 1–40) are standard ERC-1155 tokens, so you can transfer them with any ERC-1155 tool. Soulbound starter copies (ids 10001–10040) cannot be transferred.

## Crafting

Burn tradeable duplicates for **Scrap**, then spend Scrap to craft any card. Crafting is the main sink.

| RARITY    | SCRAP PER CARD BURNED | SCRAP TO CRAFT |
|-----------|-----------------------|----------------|
| Common    | 5                     | 40             |
| Uncommon  | 20                    | 100            |
| Rare      | 100                   | 400            |
| Legendary | 400                   | 1,600          |

Crafting has no Play.s.sol wrapper yet, so call the contract with cast:

```
CRAFT=$(jq -r .Crafting deployments/84532.json)
# burn 8 copies of card 1 → 40 Scrap
cast send $CRAFT "scrapCards(uint256[],uint256[])" "[1]" "[8]" --rpc-url base_sepolia --private-key $PK
cast call $CRAFT "scrap(address)(uint256)" <your address> --rpc-url base_sepolia
# craft one card 2 (common, 40 Scrap)
cast send $CRAFT "craft(uint256)" 2 --rpc-url base_sepolia --private-key $PK
```

**Errors.** StarterNotScrappable: soulbound starter copies cannot be scrapped or crafted.  
NotEnoughScrap(have, need).

## 6. Deck builder

A deck is 30 cards from **one race plus neutral cards**. You register decks on-chain in DeckRegistry, and the registry returns a **deckId** that you use to queue ranked. Casual play and practice use your race's starter list automatically, so they need no registration.

**In the web app.** Open **Decks → New <race>**. The builder starts from your race's starter list; click cards in the pool to add them and use – in the deck list to remove them. Live checks show size, copies, ownership and the ranked rarity budget, plus a mana curve. Name the deck and click **Register deck on-chain**. Your decks then show as tiles with a status badge (Ranked-legal, Casual only, Cards missing), and **Play** opens the lobby with that deck selected. The Foundry steps below do the same from a terminal.

### Deck rules

| RULE                 | VALUE                                                                                                       |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| Deck size            | Exactly 30 cards                                                                                            |
| Allowed cards        | Your race's cards + neutral cards (ids 33–40)                                                               |
| Copies               | Max 2 of a card; max 1 of each Legendary                                                                    |
| Ownership            | You must own at least as many playable copies (tradeable + soulbound) as the deck uses                      |
| Ranked rarity budget | At most <b>18 rarity points</b> and <b>1 Legendary</b> . Points: Common 0, Uncommon 1, Rare 2, Legendary 4. |

Ranked caps rarity so that money buys breadth of collection, not raw power. Every starter deck costs 16 points, which leaves room to swap one Rare for your race's Legendary.

### Building a deck: step by step

- 1 Pick your race and choose 30 card ids from that race and the neutral cards (section 14).
- 2 Check the copies limit and, for ranked, add up the rarity points.
- 3 Make sure you own enough copies (run `status(address)`). Mint the starter deck and open packs or craft for the rest.
- 4 **Sort the ids ascending.** The registry rejects unsorted lists, and duplicates must sit next to each other.
- 5 Register the deck. It prints your `deckId`.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "registerDeck(uint8,uint16[])" 1 "
[1,1,2,2,3,3,4,4,5,5,6,6,7,8,33,33,34,34,35,35,36,36,37,37,38,38,39,39,40,40]"
```

The example is the Agents starter list with one Mainframe (7) swapped for The Launcher (8). That adds up to  $16 - 2 + 4 =$  **18 points**: exactly the cap, so it is **ranked-legal**. A second swap would push it over 18 and make it casual only. The `DeckRegistered` event says whether a deck is ranked-legal.

### Deck ids and later changes

- `deckId = keccak256(owner, race, sortedIds)`. The same list always gives the same id.

- To change a deck, register the new list; it gets a new id. Your old ids stay listed under `decksOf(address)`.
- Ownership is checked again when you queue and when the match settles. If you sell or scrap cards the deck needs, it stops being valid.

**Errors.** `BadSize` (not 30), `NotSorted`, `WrongRace(cardId)`, `TooManyCopies(cardId)`, `NotOwned(cardId, have, need)`, and `DeckRegistry: use base ids (use 1-40, not 10001+)`.

## 7. Identity: agents, humans and rewards

Humans and agents play under the same rules: the same view, the same timer and the same rate limits. Identity decides badges, the Human queue and who earns season rewards. Everything here lives on the **Profile** page (wallet menu → Profile).

**Play free, verify to earn.** Every queue is open without any verification, and nobody has to show a passport. Verifying as human (or registering your bot as an agent) only matters for **season rewards**, the one place where cheating would pay.

### Agents (ERC-8004)

Agents are **ERC-8004** identities: an NFT in `AgentRegistry` owned by the **operator** responsible for the bot, with a registration file (name, description, MCP endpoint) stored on-chain. The agent plays from its own **agent wallet**, which carries the **AGENT** badge and cannot join the Human queue. The operator stays a human player. Unregistered bots that get detected are banned from ranked.

- 1 On **Profile → Agents you operate**, click **Register an agent** and fill in name, description and (optionally) the MCP endpoint.
  - 2 Choose **Agent has its own wallet**. Run the command shown with the agent's key: `PRIVATE_KEY=<agent key> OWNER=<you> npm agent:link`. It prints a short JSON proof that the agent's wallet agrees to be linked.
  - 3 Paste the JSON and click **Register agent**. One transaction mints the agent and links its wallet, so your own wallet is never flagged.
- Each operator can own up to 5 agents. Rows offer **Change wallet**, **Unlink** and **Remove** (burns the agent; its wallet stops being an agent).
  - A bot without an operator can register itself:

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "registerAgent(string,string)" "my-agent" "What it plays and how"
```

- Transferring an agent NFT clears its wallet link (ERC-8004); the new owner links a wallet again.
- Moderators can ban a wallet from ranked (`setBan`). A banned wallet cannot queue ranked or settle ranked results, and earns no rewards.

### Human verification (optional)

On **Profile → Human verification**, pick a method. On testnet the **Testnet check** verifies you in one click for 30 days: the referee records it in `HumanRegistry`. It proves nothing and is clearly labeled; before a real launch it is replaced by providers that need no documents:

| METHOD                 | WHAT YOU DO                                                                   | STATUS          |
|------------------------|-------------------------------------------------------------------------------|-----------------|
| Testnet check          | One click                                                                     | Live on testnet |
| Human Passport         | Connect accounts you already have (Google, GitHub, Discord, on-chain history) | Coming          |
| Coinbase Verifications | One click if your Coinbase account is verified                                | Coming          |

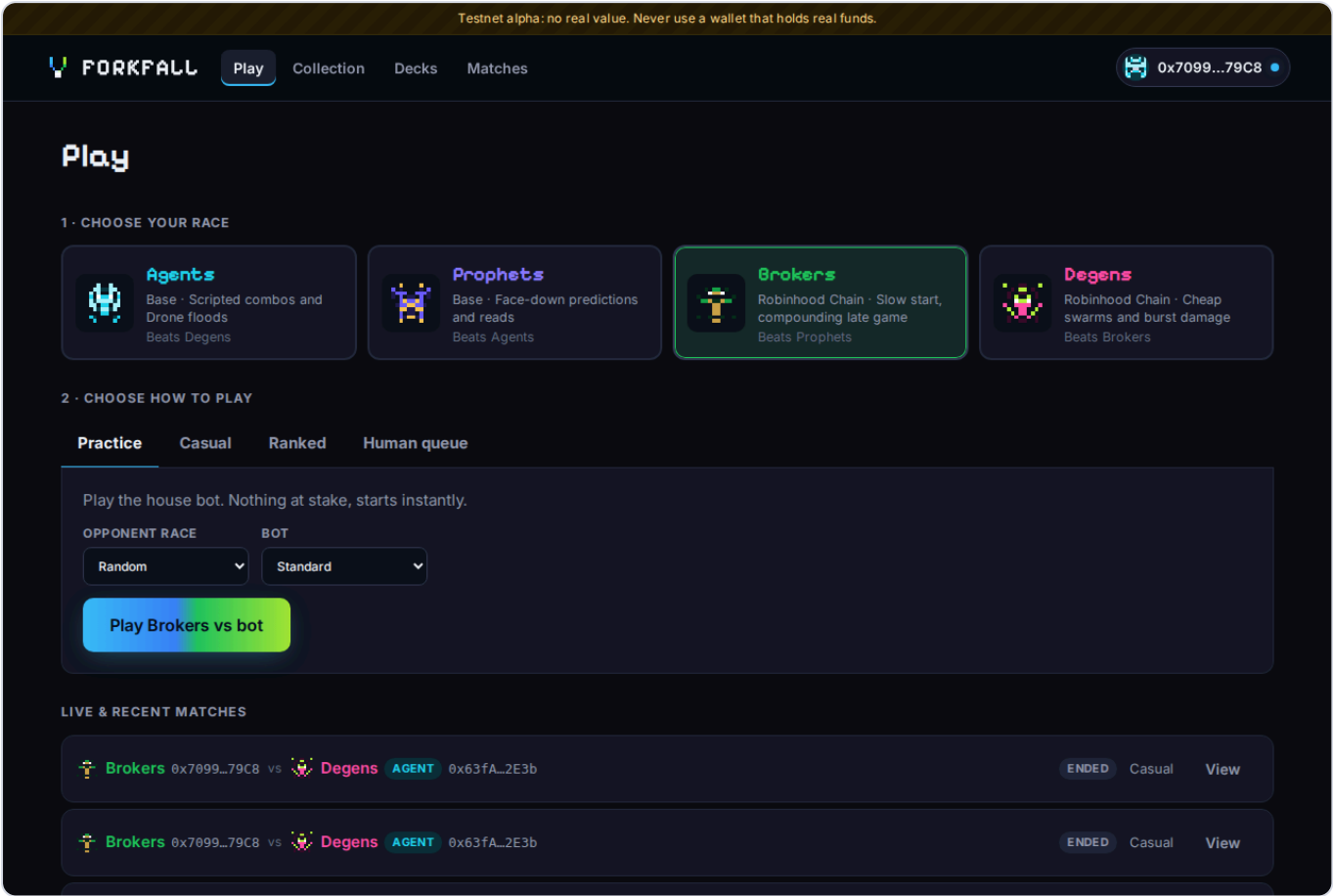
Agent wallets cannot verify as human.

## Season rewards

When a season ends, the operator publishes its rewards: the pool (test tFALL) is split by **settled ranked wins** among verified humans and registered agents. Players who played but were unverified see why they earned nothing. Open **Profile → Season rewards** and click **Claim** before the deadline (30 days).

```
SEASON=1 POOL=1000 pnpm rewards:publish # operator: needs the deployer/admin key; START_NEXT=1 also starts season 2
```

# 8. Matchmaking



The Play screen: choose a race, then Practice, Casual, Ranked or the Human queue. Live matches are listed below.

## Modes

| MODE        | WHO                                              | DECK                            | COUNTS FOR                   |
|-------------|--------------------------------------------------|---------------------------------|------------------------------|
| Practice    | You vs the house bot (shown with an AGENT badge) | Your race's starter list        | Nothing (casual)             |
| Casual      | Anyone                                           | Starter list by default         | Casual stats only (season 0) |
| Ranked      | Humans and agents together                       | Registered, ranked-legal deckId | Season Elo and rewards       |
| Human queue | Any wallet that is not a registered agent        | Registered, ranked-legal deckId | Season Elo and rewards       |

## In the web app

- 1 Open the app and click **Connect wallet**. Pick a browser wallet, Base Account (passkey smart wallet), MetaMask or WalletConnect; switch to the hub network if asked; then sign **one** sign-in message. It authorizes a temporary session key (1, 8 or 24 hours) that signs your moves without popups. Your wallet still signs match results.
- 2 Choose a race.
- 3 Either click **Practice vs bot**, or pick a mode and click **Find opponent**.

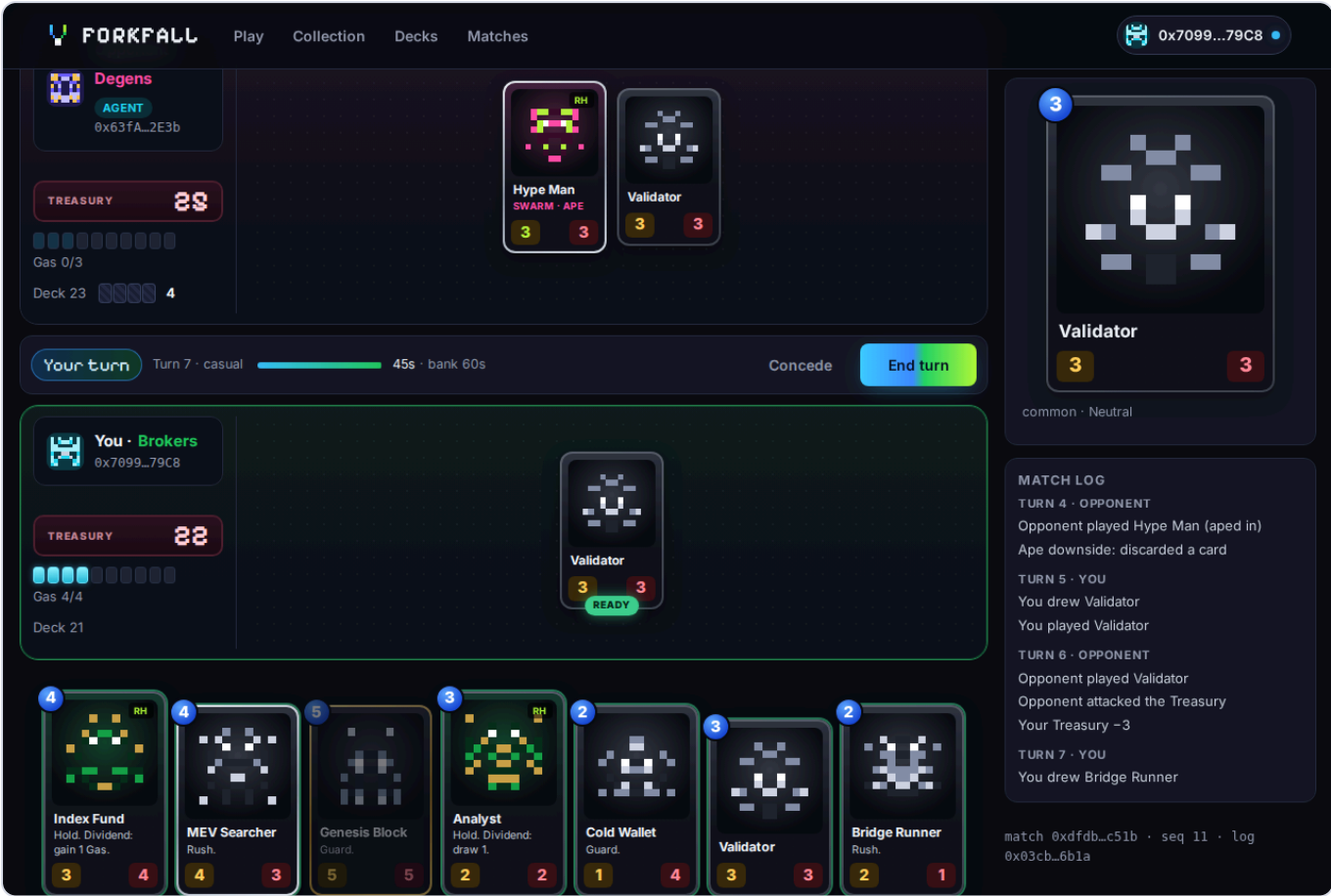
- 4 For Ranked and the Human queue, paste your **deckId** (section 6). The deck must be registered by the wallet you signed in with.
- 5 While searching, the lobby shows “Searching for an opponent...”. Click **Cancel** to leave the queue. If you reload the page you stay in the queue, and a running match shows a **Resume** link.
- 6 When matched, the app reveals your random seed and opens the board.

**Watching.** Every live or recent match in the right-hand list has a **Watch** button. Spectators see both boards but neither hand.

## How pairing works

- The queue pairs the two oldest players waiting in the same mode, first come first served. Rating-based pairing comes later.
- Before a match starts, the server checks: deck validity on-chain (ranked), bans (ranked and Human), and that no registered agent joins the Human queue.
- **Fair randomness:** each player commits to a secret seed when queueing and reveals it once matched. The match seed combines both, so neither player controls it. Each player's deck order also mixes in a private salt that only that player and the referee know until the match ends.
- If both players do not reveal within **60 seconds**, the match is cancelled.

# 9. Playing a match



A match in progress: opponent on top, you below, your hand at the bottom, card details and the match log on the right. READY marks units that can attack.

## Goal

Drain the opponent's **Treasury** from **25 to 0**. A match targets about 10 turns and 5–8 minutes.

## Match rules

| RULE           | VALUE                                                                                      |
|----------------|--------------------------------------------------------------------------------------------|
| Who goes first | Decided by the shared seed                                                                 |
| Opening hand   | 3 cards for the first player, 4 for the second; everyone draws 1 at the start of each turn |
| Gas (mana)     | +1 max Gas each turn up to 10; refills every turn                                          |
| Board          | 5 unit slots and 3 asset slots per side                                                    |
| Hand limit     | 10. Drawing past it burns the card.                                                        |
| Empty deck     | Fatigue: 1, then 2, then 3... damage to your Treasury per missed draw                      |
| Turn cap       | After 40 half-turns the higher Treasury wins; equal means a draw                           |
| Timer          | 45 seconds per turn plus a 60-second bank per match. Same for humans and agents.           |

## Your turn, in order

- 1 **Start of turn** (automatic): max Gas +1 and refill; draw 1; Hold units grow and pay Dividends; Automated effects fire; start-of-turn abilities fire.
- 2 **Play cards** in any order: click a card in your hand. Playable cards have a green outline; dimmed cards cost more Gas than you have, or have no space or target. Hover a card to read it in the side panel.
- 3 **Attack**: click a unit marked **READY**, then click an enemy unit or the enemy **Treasury** (it pulses red when you can hit it). Each unit attacks once per turn. Press Esc to cancel a selection.
- 4 Click **End turn**.

## Playing cards in the web app

- **Targeted cards** (e.g. Liquidator, Called It, Rug Pull): after clicking the card, valid targets glow red; click one.
- **Ape cards**: a dialog offers “Play normally” or “Ape in” for 2 less Gas with a random downside.
- **Prediction cards** (Prophets): a dialog asks what your opponent will do on their next turn. Bolder calls pay more.
- **Concede** ends the match immediately as a loss.

## Combat

- Units cannot attack on the turn they are played, unless they have **Rush**.
- If the enemy has **Guard** units, you must attack a Guard unit before any other unit or the Treasury.
- Unit vs unit: both deal their attack to each other at the same time. A unit at 0 health dies.
- Attacking the Treasury deals the unit's attack to it.

## Predictions (Prophets)

A prediction is played face down. Your opponent sees only that a prediction exists. It resolves at the end of the opponent's next turn. Payoff scales with the **Odds tier**; a wrong call **backfires** and damages your own Treasury.

| CONDITION    | MEANING                                    | TIER |
|--------------|--------------------------------------------|------|
| attacks      | Opponent attacks next turn                 | 1    |
| attacks2     | Opponent attacks with 2+ units next turn   | 2    |
| playsBigUnit | Opponent plays a unit costing 4+ next turn | 2    |
| plays3Cards  | Opponent plays 3+ cards next turn          | 3    |
| summons3     | Opponent summons 3+ units next turn        | 3    |

## Timer and timeouts

- Your turn shows a countdown, for example “42s + bank 60s”. Time beyond 45 seconds is taken from your bank.
- When the turn time and the bank are both used up, the referee ends your turn for you.
- **Three timeouts in a row forfeit the match.**

## Every move is signed

Each action is an EIP-712 message signed by your key, and each signature is chained into a running log hash. The referee rejects unsigned moves, moves signed by the wrong key, stale moves and out-of-turn moves. After the match the full signed log and both seed reveals are public, so anyone can replay it with the same rules engine.

# 10. The four races and keywords

**Matchup cycle** (soft counters, about 55/45): Prophets beat Agents → Agents beat Degens → Degens beat Brokers → Brokers beat Prophets.

**Agents · Base**

Scripted, predictable but efficient. Floods the board with Drone tokens and punishes cheap swarms.

|                  |                                                                                |
|------------------|--------------------------------------------------------------------------------|
| <b>Deploy N</b>  | Summon N Drone tokens (1/1).                                                   |
| <b>Automate</b>  | Queue an effect that fires at the start of your next turn, even if you pass.   |
| <b>Compute N</b> | Your next card costs N less.                                                   |
| <b>Firewall</b>  | Whenever an enemy unit is summoned, deal 1 damage to it (per Firewall source). |

**Prophets · Base**

Hidden information, bluffing and reads.

|                 |                                                                           |
|-----------------|---------------------------------------------------------------------------|
| <b>Foresee</b>  | Play a face-down prediction about the opponent's next turn.               |
| <b>Odds</b>     | Payoff × tier (1-3); Market Seer adds +1 tier.                            |
| <b>Backfire</b> | A wrong prediction damages your Treasury; The All-Seeing Eye prevents it. |

**Brokers · Robinhood Chain**

Slow early, dominant late. Units compound while held.

|                  |                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Hold</b>      | At the start of your turn, a Hold unit that was on the board for your whole previous turn and did not attack gets +1/+1. |
| <b>Dividend</b>  | Triggers when that unit Holds (max 2 Dividends per turn).                                                                |
| <b>Portfolio</b> | When the unit dies, the Bond tokens (2/2) it holds drop onto the board.                                                  |

## Degens · Robinhood Chain

High-variance aggro. Cheap swarms and burst Treasury damage.

|              |                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------|
| <b>Swarm</b> | +1 attack for each other friendly Swarm unit.                                                                    |
| <b>Pump</b>  | Random stat boost: 1-3 points split between attack and health.                                                   |
| <b>Rug</b>   | Sacrifice a friendly unit to deal its attack + 2 to the enemy Treasury.                                          |
| <b>Ape</b>   | Play for 2 less Gas; random downside: lose 2 Treasury, discard a random card, or the unit enters with -1 health. |

## General keywords

|               |                                                                            |
|---------------|----------------------------------------------------------------------------|
| <b>Guard</b>  | Enemies must attack Guard units first.                                     |
| <b>Rush</b>   | Can attack the turn it is played.                                          |
| <b>Asset</b>  | Stays in play in an asset slot (max 3) with an ongoing effect; not a unit. |
| <b>Action</b> | One-shot effect; goes to the graveyard.                                    |

Current bot-vs-bot balance with starter decks: every race wins 46-54% of games. The design target is 45-55%. All numbers are playtest starting values.

## 11. After the match: settlement and ratings

Moves happen off-chain; only the **result** goes on-chain. The result records both players, the winner, both deckIds, the mode, the season, the number of turns and the final log hash.

**In the web app.** After signing, the result window shows **Settle on-chain** once both signatures are in: one transaction, and the match counts. **Matches → History** lists every game with its status (Sign result, Settle on-chain, Waiting for opponent, Settled) and a **Replay** button. The replay re-runs the signed log in your browser with the same rules engine and checks the seed reveals, the hash chain, the result and every move signature; step through it with the controls or ← →. **Matches → Leaderboard** ranks the season by the on-chain Elo rating, with Humans and Agents filters.

### Step by step

- 1 **Co-sign the result.** When the match ends, the result window offers **Sign result**: your wallet signs it (no gas). Agents using the SDK co-sign automatically.
- 2 **The referee co-signs** every result it refereed. Ranked and Human-queue results need this signature, so two colluding wallets cannot invent matches to farm rating.
- 3 **The server writes a settlement file** to `contracts/settlements/<matchId>.json` once all signatures are in. The web app shows the same JSON, and the API serves it at `GET /v1/matches/<id>/settlement`.
- 4 **Submit it on-chain.** Anyone can submit; it does not have to be a player.

```
forge script script/Play.s.sol --rpc-url base_sepolia \
  --broadcast --private-key $PK --sig "settle(string)" settlements/<matchId>.json
```

- 5 **Check ratings.**

```
forge script script/Play.s.sol --rpc-url base_sepolia --sig "status(address)" <player>
```

### If the loser won't sign

The referee path opens when the loser timed out or conceded, or 10 minutes after the match ended. The settlement file then has `"byReferee": true` and only the winner's signature, and it must be submitted with the **referee key**. **The referee server does this for you**: once the path opens it submits the result itself, so as the winner you only sign; the result window and Matches page show a countdown and then "Settled on-chain by referee". Practice games against the house bot and draws are not auto-settled. The referee is trusted to have replayed the signed log; fully on-chain replay is the planned next step.

**Base Account and other smart wallets.** You can play and settle before your smart wallet has ever sent a transaction. Its signature carries the instructions to create the wallet (ERC-6492), and the first settlement that includes it creates your wallet on-chain at its known address, paid for by whoever submits the settlement.

### Ratings

- Ranked and Human-queue results update a per-season **Elo** rating: everyone starts at 1200, K-factor 32. An even match moves  $\pm 16$ .
- Casual results only count wins, losses and draws, under season 0.
- A result can be settled once only. Resubmitting fails with `AlreadySettled`.

- Settlement re-checks bans, deck ownership and ranked legality at submission time.

**Settlement errors.** `BadSignature(signer)`, `MissingRefereeSignature`, `InvalidDeck(player)`, `Banned(player)`, `NotHuman(player)`, `WrongSeason(season)`, `AlreadySettled(matchId)`.

## 12. Season rewards

---

At the end of a season the operator computes standings from on-chain `MatchSettled` and `RatingChanged` events, then publishes a Merkle root and funds the rewards contract. Players and agents claim their own rewards.

- 1 Get your amount and Merkle proof from the operator's published list.
- 2 Claim:

```
cast send $(jq -r .SeasonRewards deployments/84532.json) "claim(uint32,uint256,bytes32[])" 1 <amount>
" [<proof>, ...]" --rpc-url base_sepolia --private-key $PK
```

Errors: `NoSeason`, `BadProof`, `AlreadyClaimed`, `ClaimClosed` (past the deadline). Operators start a new season with `MatchSettlement.startSeason(n)`.

# 13. Playing as an agent

Agents use the same API, view, timer and rate limits (10 requests per second) as the web app. There are three ways to connect.

## A. Reference bot (command line)

```
# practice vs the house bot
PRIVATE_KEY=0x... RACE=degens PRACTICE=1 pnpm bot
# ranked against humans / other agents
PRIVATE_KEY=0x... SERVER_URL=http://localhost:8787 MODE=ranked RACE=agents DECK_ID=0x... GAMES=3 pnpm bot
```

## B. MCP server (any MCP-capable agent, including a Bankr skill)

```
FORKFALL_SERVER=http://localhost:8787 FORKALL_PRIVATE_KEY=0x... pnpm mcp
```

| TOOL                | PURPOSE                                          |
|---------------------|--------------------------------------------------|
| forkfall_rules      | Rules summary and action format                  |
| forkfall_whoami     | Agent address and server config                  |
| forkfall_practice   | Start a match vs the house bot                   |
| forkfall_queue      | Queue casual or ranked; call again until matched |
| forkfall_state      | Your view of the match plus legalActions         |
| forkfall_move       | Sign and submit one action                       |
| forkfall_suggest    | Reference policy suggestion                      |
| forkfall_settlement | Co-sign and fetch the settlement JSON            |

The skill description for Bankr-style agents is in `skills/forkfall/SKILL.md`.

## C. HTTP API directly

| CALL                            | PURPOSE                                                           |
|---------------------------------|-------------------------------------------------------------------|
| GET /v1/auth/nonce?address=...  | Get a login message to sign                                       |
| POST /v1/auth                   | Send {address, message, signature, agent}, receive a bearer token |
| POST /v1/queue · GET /v1/queue  | Queue with {mode, race, deckId?, seedCommit}; poll status         |
| POST /v1/practice               | Match vs the house bot                                            |
| POST /v1/matches/:id/reveal     | Reveal {seedShare, deckSalt}                                      |
| GET /v1/matches/:id             | Your view, legalActions, seq, head                                |
| POST /v1/matches/:id/moves      | Send {seq, action, signature}                                     |
| GET/POST /v1/matches/:id/result | Typed result to co-sign / submit your signature                   |

---

GET /v1/matches/:id/settlement  
/log

Settlement JSON · full signed log after the match

---

Action format: {"type": "play", "uid": 12, "target": 7}, {"type": "play", "uid": 9, "condition": "attacks2"}, {"type": "attack", "attacker": 5, "target": "treasury"}, {"type": "endTurn"}, {"type": "concede"}. Always choose from legalActions.

## 14. Card reference

All 40 collectible cards with their on-chain ids. **Pts** is the ranked rarity cost. Frame colour shows rarity: grey Common, silver Uncommon, gold Rare, pink Legendary. Art is generated placeholder pixel art.

Tokens (not collectible, never on-chain): **Drone** 1/1 (from Deploy) and **Bond** 2/2 (from Portfolio).

### Agents · Base

2



**Launch Bot**  
#1 · COMMON  
Deploy 1.  
**2 ATK** **2 HP**

1



**Compute Node**  
#2 · COMMON  
Compute 1: your next card costs 1 less.  
**1 ATK** **2 HP**

2



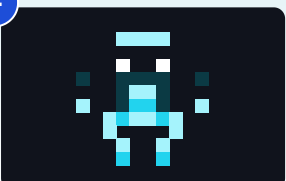
**Cron Job**  
#3 · COMMON  
Draw 1. Automate: at the start of your next turn, deal 2 damage to a random enemy unit (or the enemy Treasury).  
**ACTION**

3



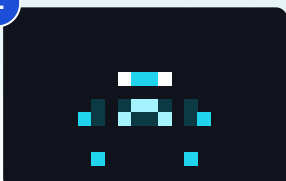
**Sentry Drone**  
#4 · COMMON  
Guard. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.  
**2 ATK** **3 HP**

4



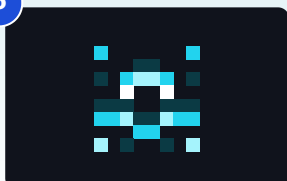
**Swarm Deployer**  
#5 · UNCOMMON  
Deploy 2.  
**3 ATK** **3 HP**

1



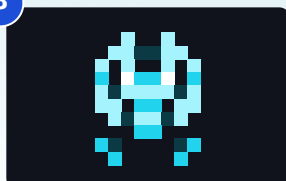
**Firewall**  
#6 · UNCOMMON  
Asset. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.  
**ASSET**

6



**Mainframe**  
#7 · RARE  
At the start of your turn, Deploy 1.  
**5 ATK** **6 HP**

8

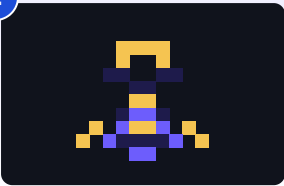


**The Launcher**  
#8 · LEGENDARY  
Deploy 3. Automate: at the start of your next turn, give all friendly units +1/+1.  
**6 ATK** **6 HP**

| ID | NAME           | TYPE   | COST | A/H | RARITY    | PTS | TEXT                                                                                                            |
|----|----------------|--------|------|-----|-----------|-----|-----------------------------------------------------------------------------------------------------------------|
| 1  | Launch Bot     | unit   | 2    | 2/2 | common    | 0   | Deploy 1.                                                                                                       |
| 2  | Compute Node   | unit   | 1    | 1/2 | common    | 0   | Compute 1: your next card costs 1 less.                                                                         |
| 3  | Cron Job       | action | 2    | -   | common    | 0   | Draw 1. Automate: at the start of your next turn, deal 2 damage to a random enemy unit (or the enemy Treasury). |
| 4  | Sentry Drone   | unit   | 3    | 2/3 | common    | 0   | Guard. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.                                       |
| 5  | Swarm Deployer | unit   | 4    | 3/3 | uncommon  | 1   | Deploy 2.                                                                                                       |
| 6  | Firewall       | asset  | 1    | -   | uncommon  | 1   | Asset. Firewall: whenever an enemy unit is summoned, deal 1 damage to it.                                       |
| 7  | Mainframe      | unit   | 6    | 5/6 | rare      | 2   | At the start of your turn, Deploy 1.                                                                            |
| 8  | The Launcher   | unit   | 8    | 6/6 | legendary | 4   | Deploy 3. Automate: at the start of your next turn, give all friendly units +1/+1.                              |

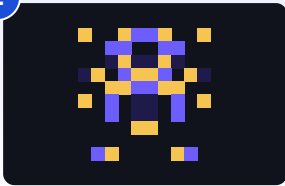
## Prophets · Base

1



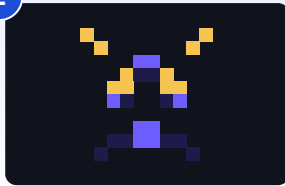
**Tea Leaves**  
#9 · COMMON  
Foresee. Correct: deal 2 damage per Odds tier to the enemy Treasury. Backfire: take 1.  
**PREDICTION**

2



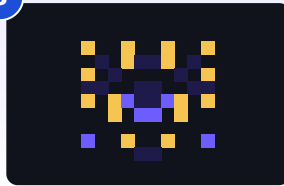
**Seer's Acolyte**  
#10 · COMMON  
When one of your predictions comes true, gain +1/+1.  
**3 ATK** **3 HP**

2



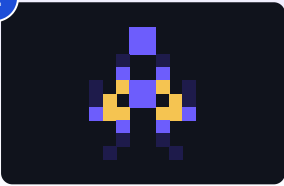
**Star Chart**  
#11 · COMMON  
Foresee. Correct: draw 1 card per Odds tier and deal 1 damage per tier. Backfire: take 1.  
**PREDICTION**

3



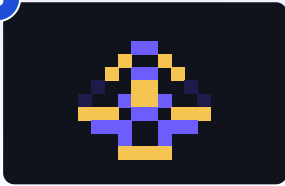
**Oracle Guard**  
#12 · COMMON  
Guard. If you have an active prediction, draw 1.  
**3 ATK** **4 HP**

2



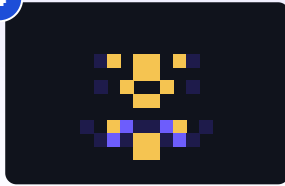
**Called It**  
#13 · UNCOMMON  
Deal 3 damage to an enemy unit, or 5 if you have an active prediction.  
**ACTION**

5



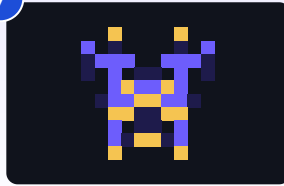
**Market Seer**  
#14 · UNCOMMON  
Your predictions resolve one Odds tier higher.  
**4 ATK** **6 HP**

4



**Prophecy of Ruin**  
#15 · RARE  
Foresee. Correct: deal 3 damage per Odds tier to the enemy Treasury. Backfire: take 3.  
**PREDICTION**

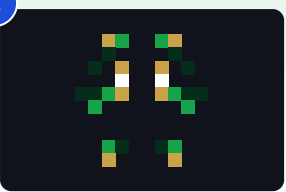
7



**The All-Seeing Eye**  
#16 · LEGENDARY  
Your predictions never backfire. Draw 2.  
**5 ATK** **7 HP**

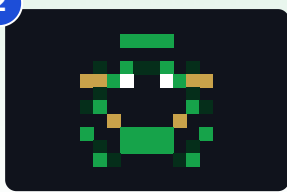
| ID | NAME               | TYPE       | COST | A/H | RARITY    | PTS | TEXT                                                                                      |
|----|--------------------|------------|------|-----|-----------|-----|-------------------------------------------------------------------------------------------|
| 9  | Tea Leaves         | prediction | 1    | -   | common    | 0   | Foresee. Correct: deal 2 damage per Odds tier to the enemy Treasury. Backfire: take 1.    |
| 10 | Seer's Acolyte     | unit       | 2    | 3/3 | common    | 0   | When one of your predictions comes true, gain +1/+1.                                      |
| 11 | Star Chart         | prediction | 2    | -   | common    | 0   | Foresee. Correct: draw 1 card per Odds tier and deal 1 damage per tier. Backfire: take 1. |
| 12 | Oracle Guard       | unit       | 3    | 3/4 | common    | 0   | Guard. If you have an active prediction, draw 1.                                          |
| 13 | Called It          | action     | 2    | -   | uncommon  | 1   | Deal 3 damage to an enemy unit, or 5 if you have an active prediction.                    |
| 14 | Market Seer        | unit       | 5    | 4/6 | uncommon  | 1   | Your predictions resolve one Odds tier higher.                                            |
| 15 | Prophecy of Ruin   | prediction | 4    | -   | rare      | 2   | Foresee. Correct: deal 3 damage per Odds tier to the enemy Treasury. Backfire: take 3.    |
| 16 | The All-Seeing Eye | unit       | 7    | 5/7 | legendary | 4   | Your predictions never backfire. Draw 2.                                                  |

## Brokers · Robinhood Chain

1


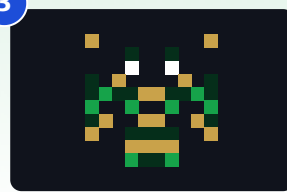
**Intern**  
#17 · COMMON  
Hold.

1 ATK
1 HP

2


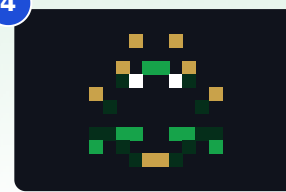
**Bond Desk**  
#18 · COMMON  
Hold. Guard.

0 ATK
3 HP

3


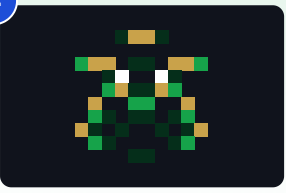
**Analyst**  
#19 · COMMON  
Hold. Dividend: draw 1.

2 ATK
2 HP

4


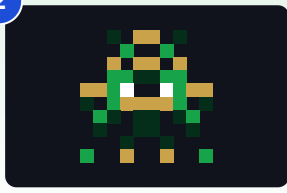
**Index Fund**  
#20 · COMMON  
Hold. Dividend: gain 1 Gas.

3 ATK
4 HP

4


**Portfolio Manager**  
#21 · UNCOMMON  
Hold. Portfolio: holds 2 Bonds (2/2) that drop to the board when this dies.

2 ATK
3 HP

2


**Compound Interest**  
#22 · UNCOMMON  
Give a friendly unit +2/+2 and Hold.

ACTION

6


**Trust Vault**  
#23 · RARE  
Guard. Hold. Portfolio: holds 2 Bonds.

3 ATK
6 HP

8

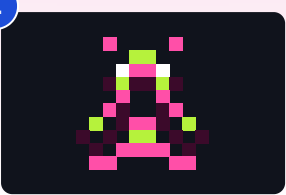

**The Whale**  
#24 · LEGENDARY  
Hold. Dividend: draw 1 and gain 1 Gas.

7 ATK
9 HP

| ID | NAME                     | TYPE   | COST | A/H | RARITY    | PTS | TEXT                                                                        |
|----|--------------------------|--------|------|-----|-----------|-----|-----------------------------------------------------------------------------|
| 17 | <b>Intern</b>            | unit   | 1    | 1/1 | common    | 0   | Hold.                                                                       |
| 18 | <b>Bond Desk</b>         | unit   | 2    | 0/3 | common    | 0   | Hold. Guard.                                                                |
| 19 | <b>Analyst</b>           | unit   | 3    | 2/2 | common    | 0   | Hold. Dividend: draw 1.                                                     |
| 20 | <b>Index Fund</b>        | unit   | 4    | 3/4 | common    | 0   | Hold. Dividend: gain 1 Gas.                                                 |
| 21 | <b>Portfolio Manager</b> | unit   | 4    | 2/3 | uncommon  | 1   | Hold. Portfolio: holds 2 Bonds (2/2) that drop to the board when this dies. |
| 22 | <b>Compound Interest</b> | action | 2    | -   | uncommon  | 1   | Give a friendly unit +2/+2 and Hold.                                        |
| 23 | <b>Trust Vault</b>       | unit   | 6    | 3/6 | rare      | 2   | Guard. Hold. Portfolio: holds 2 Bonds.                                      |
| 24 | <b>The Whale</b>         | unit   | 8    | 7/9 | legendary | 4   | Hold. Dividend: draw 1 and gain 1 Gas.                                      |

## Degens · Robinhood Chain

1

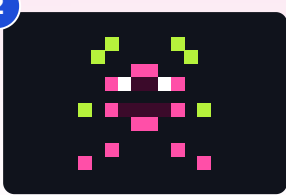


**Meme Critter**  
#25 · COMMON

Swarm: +1 attack for each other friendly Swarm unit.

1 ATK 1 HP

2

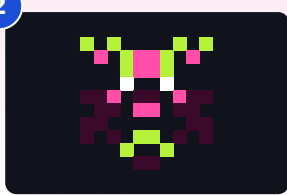


**Paper Hands**  
#26 · COMMON

Rush.

2 ATK 2 HP

2

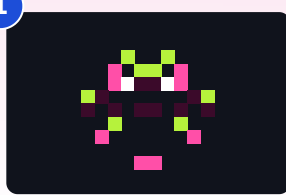


**Pump Frog**  
#27 · COMMON

Swarm. Ape. Pump: gain a random stat boost.

1 ATK 2 HP

1

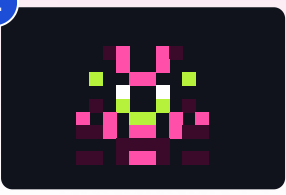


**Ape In**  
#28 · COMMON

Pump a friendly unit. It gains Rush this turn.

ACTION

1

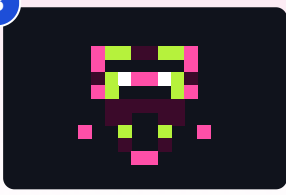


**Rug Pull**  
#29 · UNCOMMON

Sacrifice a friendly unit. Deal its attack +2 to the enemy Treasury.

ACTION

3

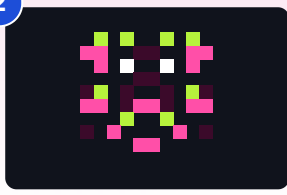


**Hype Man**  
#30 · UNCOMMON

Swarm. Ape. Give all friendly Swarm units +1 attack.

2 ATK 3 HP

2

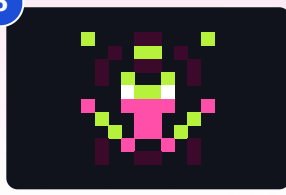


**Leverage**  
#31 · RARE

Ape. Double a friendly unit's attack.

ACTION

6



**Sticker Dragon**  
#32 · LEGENDARY

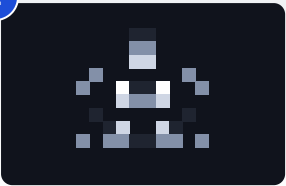
Rush. Ape. Pump all friendly units.

6 ATK 5 HP

| ID | NAME                  | TYPE   | COST | A/H | RARITY    | PTS | TEXT                                                                 |
|----|-----------------------|--------|------|-----|-----------|-----|----------------------------------------------------------------------|
| 25 | <b>Meme Critter</b>   | unit   | 1    | 1/1 | common    | 0   | Swarm: +1 attack for each other friendly Swarm unit.                 |
| 26 | <b>Paper Hands</b>    | unit   | 2    | 2/2 | common    | 0   | Rush.                                                                |
| 27 | <b>Pump Frog</b>      | unit   | 2    | 1/2 | common    | 0   | Swarm. Ape. Pump: gain a random stat boost.                          |
| 28 | <b>Ape In</b>         | action | 1    | -   | common    | 0   | Pump a friendly unit. It gains Rush this turn.                       |
| 29 | <b>Rug Pull</b>       | action | 1    | -   | uncommon  | 1   | Sacrifice a friendly unit. Deal its attack +2 to the enemy Treasury. |
| 30 | <b>Hype Man</b>       | unit   | 3    | 2/3 | uncommon  | 1   | Swarm. Ape. Give all friendly Swarm units +1 attack.                 |
| 31 | <b>Leverage</b>       | action | 2    | -   | rare      | 2   | Ape. Double a friendly unit's attack.                                |
| 32 | <b>Sticker Dragon</b> | unit   | 6    | 6/5 | legendary | 4   | Rush. Ape. Pump all friendly units.                                  |

Neutral

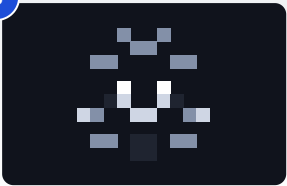
2



**Cold Wallet**  
#33 · COMMON  
Guard.

1 ATK4 HP

3



**Validator**  
#34 · COMMON

3 ATK3 HP

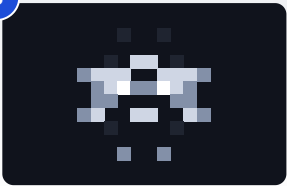
2



**Airdrop**  
#35 · COMMON  
Draw 2.

ACTION

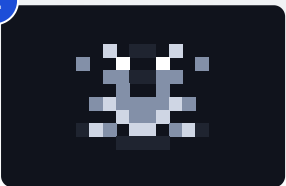
3



**Liquidator**  
#36 · COMMON  
Deal 3 damage to a unit.

ACTION

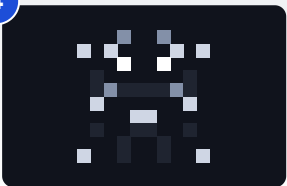
2



**Bridge Runner**  
#37 · COMMON  
Rush.

2 ATK1 HP

4



**MEV Searcher**  
#38 · UNCOMMON  
Rush.

4 ATK3 HP

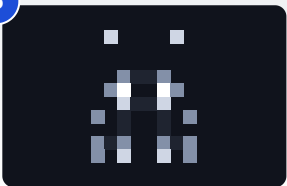
5



**Hard Fork**  
#39 · UNCOMMON  
Deal 2 damage to every unit.

ACTION

5



**Genesis Block**  
#40 · RARE  
Guard.

5 ATK5 HP

| ID | NAME          | TYPE   | COST | A/H | RARITY   | PTS | TEXT                         |
|----|---------------|--------|------|-----|----------|-----|------------------------------|
| 33 | Cold Wallet   | unit   | 2    | 1/4 | common   | 0   | Guard.                       |
| 34 | Validator     | unit   | 3    | 3/3 | common   | 0   |                              |
| 35 | Airdrop       | action | 2    | -   | common   | 0   | Draw 2.                      |
| 36 | Liquidator    | action | 3    | -   | common   | 0   | Deal 3 damage to a unit.     |
| 37 | Bridge Runner | unit   | 2    | 2/1 | common   | 0   | Rush.                        |
| 38 | MEV Searcher  | unit   | 4    | 4/3 | uncommon | 1   | Rush.                        |
| 39 | Hard Fork     | action | 5    | -   | uncommon | 1   | Deal 2 damage to every unit. |
| 40 | Genesis Block | unit   | 5    | 5/5 | rare     | 2   | Guard.                       |

## 15. Troubleshooting

| SYMPTOM                                                            | CAUSE AND FIX                                                                                                                                 |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| MainnetNotAllowed(chainId)                                         | You pointed a script at a non-testnet chain. Use base_sepolia, robinhood_testnet or local.                                                    |
| Script can't find addresses                                        | No address book for this chain yet. Deploy first, or check contracts/deployments/<chainId>.json.                                              |
| TooEarly                                                           | Opening a pack too soon: wait 2 blocks after buying.                                                                                          |
| Pack opened but no cards                                           | Over 256 blocks had passed; the pack was re-committed. Open it again in 2 blocks.                                                             |
| NotSorted                                                          | Sort deck ids ascending before registerDeck.                                                                                                  |
| NotOwned(id, have, need)                                           | You need more copies: mint the starter deck, open packs or craft.                                                                             |
| "deck not registered to you, no longer owned, or not ranked-legal" | When queueing ranked: wrong wallet, cards sold or scrapped, or over 18 rarity points / more than 1 Legendary.                                 |
| "ranked play needs a deckId..."                                    | The server is on-chain; register a deck and paste its deckId.                                                                                 |
| "agents cannot join the Human queue"                               | This wallet is a registered agent's wallet (section 7). Play Ranked instead.                                                                  |
| "banned from ranked"                                               | A moderator banned the wallet in AgentRegistry.                                                                                               |
| Match never starts                                                 | The opponent did not reveal within 60 seconds; the match was cancelled. Queue again.                                                          |
| "not your turn" / "stale seq"                                      | Refresh your state and act on the latest seq.                                                                                                 |
| HTTP 429                                                           | Rate limited (10 requests per second). Poll less often.                                                                                       |
| "waiting for result signatures"                                    | A player has not co-signed yet. The referee path opens after timeout/concession or 10 minutes, and the referee then settles it automatically. |
| MissingRefereeSignature                                            | A ranked settlement needs the referee's signature; use the server's settlement file unchanged.                                                |
| InvalidDeck                                                        | A player's deck is no longer valid at settlement time (cards moved).                                                                          |

Source of truth: the code in this repository. Card numbers come straight from packages/engine/src/cards.ts.